

Ski Rental Problem

1 Problem

Ski Rental

A skier does not know how many days they will ski. Each day, before knowing the future, the skier must choose one of two actions:

rent for one day or buy once and use forever.

Use the values

$$R = 500, \quad B = 1500,$$

where R is the rental cost per day and B is the buying cost.

Let d be the actual number of ski days.

2 Offline Optimum

If d were known in advance, the best offline strategy would be:

$$\text{OPT}(d) = \min(Rd, B).$$

For our values,

$$\text{OPT}(d) = \min(500d, 1500).$$

Meaning

The offline optimum either rents for all d days or buys immediately. It chooses whichever is cheaper after seeing the whole future.

3 Break-Even Day

The break-even point is the number of rental days whose total cost equals buying:

$$\frac{B}{R} = \frac{1500}{500} = 3.$$

So:

$$3 \text{ rental days cost } 3 \cdot 500 = 1500,$$

which is exactly the cost of buying.



4 Bad Strategy: Always Rent

Cost

$$C_{\text{rent}}(d) = 500d.$$

If $d \leq 3$, this is fine. But if d becomes large, the offline optimum buys once and pays only 1500. For example, if $d = 10$,

$$C_{\text{rent}}(10) = 5000, \quad \text{OPT}(10) = 1500.$$

The ratio is

$$\frac{5000}{1500} = \frac{10}{3}.$$

As d grows,

$$\frac{500d}{1500} = \frac{d}{3}$$

also grows without bound.

Conclusion

Always renting is not competitive because its competitive ratio is unbounded.

5 Bad Strategy: Buy Immediately

Cost

$$C_{\text{buy-now}}(d) = 1500.$$

This is good when the skier skis many days, but bad when the skier stops early.

If $d = 1$, then

$$C_{\text{buy-now}}(1) = 1500, \quad \text{OPT}(1) = 500.$$

So

$$\frac{C_{\text{buy-now}}(1)}{\text{OPT}(1)} = \frac{1500}{500} = 3.$$

Conclusion

Buying immediately is 3-competitive.

6 Good Strategy: Rent Twice, Then Buy

Since the break-even point is day 3, use this deterministic online strategy:

Threshold Strategy

Rent on days 1 and 2. If skiing continues to day 3, buy at the start of day 3.

The online cost is

$$C_A(d) = \begin{cases} 500d, & d \leq 2, \\ 2500, & d \geq 3. \end{cases}$$

The value 2500 comes from

$$2 \cdot 500 + 1500 = 2500.$$

7 Why the Ratio Is $\frac{5}{3}$

We prove

$$C_A(d) \leq \frac{5}{3} \text{OPT}(d)$$

for every possible number of days d .

Case 1: $d \leq 2$

The algorithm rents for every ski day:

$$C_A(d) = 500d.$$

The offline optimum also rents:

$$\text{OPT}(d) = 500d.$$

Thus

$$\frac{C_A(d)}{\text{OPT}(d)} = 1.$$

Since

$$1 \leq \frac{5}{3},$$

the bound holds.

Case 2: $d \geq 3$

The algorithm rents twice and then buys:

$$C_A(d) = 2500.$$

The offline optimum buys, because renting 3 or more days costs at least 1500:

$$\text{OPT}(d) = 1500.$$

Therefore

$$\frac{C_A(d)}{\text{OPT}(d)} = \frac{2500}{1500} = \frac{5}{3}.$$

Competitive Ratio

Rent twice, then buy is $\frac{5}{3}$ -competitive.

8 Cost Table

Days d	Online $C_A(d)$	Offline $\text{OPT}(d)$	Ratio
1	500	500	1
2	1000	1000	1
3	2500	1500	$\frac{5}{3}$
4	2500	1500	$\frac{5}{3}$
10	2500	1500	$\frac{5}{3}$

The worst ratio starts when the skier reaches day 3.

9 General Rule

Let:

R = daily rental cost, B = buying cost.

If B/R is an integer, then the standard deterministic threshold strategy is:

rent for $\frac{B}{R} - 1$ days, then buy on the next day.

Its competitive ratio is

$$2 - \frac{R}{B}.$$

With the given values,

$$2 - \frac{R}{B} = 2 - \frac{500}{1500} = 2 - \frac{1}{3} = \frac{5}{3}.$$

10 Strategy Comparison

Strategy	Worst ratio	Problem
Always rent	Unbounded	Very expensive if the season is long.
Buy immediately	3	Too expensive if the skier stops after one day.
Rent twice, then buy	$\frac{5}{3}$	Balances early stopping and long seasons.

11 Review Box

Core Facts

$$\text{OPT}(d) = \min(Rd, B).$$

$$\text{Break-even day} = B/R.$$

For $R = 500$, $B = 1500$, break-even day = 3.

Best simple deterministic strategy: rent days 1, 2, buy on day 3.

$$\text{Competitive ratio} = \frac{2500}{1500} = \frac{5}{3}.$$